MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, OSCILLATOR, QUINARY, MEMORY AND TENS MATRIX - A1: 10547210

1. SCOPE

1.1 This specification covers a multi-layer printed circuit board, a sub-assembly of the counter, Time Interval: 10547200 which is a modular component of the Range Finder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Printed Circuit Assemblies
MIL-P-46843	Printed Circuit Assemblies, Design and Procurement of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10547210	Circuit Card Assembly - Oscillator, Quinary, Memory and Tens Matrix - A1

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PACKAGING DATA SHEETS

MIL-P-14232 10547210	Packaging of Circuit Card Assembly - Oscillator, Quinary, Memory and Tens Matrix - A1

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The Circuit Card Assembly - Oscillator, Quinary, Memory and Tens Matrix - A1, hereinafter referred to as assembly, shall be manufactured in accordance with Drawing F10547210 and drawings pertaining thereto.

3.3 General specification.- The assembly shall meet the following requirements, where applicable, of MIL-F-13926 and MIL-P-46843.

a. Dimensions and tolerances
b. Inorganic protection surface finisher
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926 and MIL-P-46843, the assembly shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.11.4 inclusive.

3.3.1.1 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface. Components stacked directly above another component shall be mounted with at least one point of the upper component in contact with the bottom component. The upper component shall not be raised more than 0.031 inch from the bottom component at any point on its contacting surface.

3.3.1.2 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that the proper identification markings on each part are visible.

3.3.1.3 Transistors.- Transistors are to be mounted perpendicular to the circuit board with spacers between the transistor and the circuit board.

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3.3.1.4 Bend radius.- The minimum bend radii for formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.1.5 Component leads.- The minimum lead length for radial lead components shall be 4 times the lead diameter. The portion of the lead between a bend and the component body shall emerge from the body in a straight line radially from the component body and enter the plated hole approximately perpendicular to the board. Unless otherwise specified in the applicable drawing, leads of transistors mounted with a spacer directly into the printed circuit board shall have a minimum of .062 inch of lead from the transistor base to any soldered joint. Component leads shall be pre-formed to suit mounting holes, and clinched parallel to board after being inserted through the holes. There shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than .060 inch, including solder, beyond the side of the circuit board opposite the component. The leads shall not extend more than two lead diameters beyond the land area.

3.3.1.6 Soldering.- All solder joints shall conform to MIL-F-13926 and MIL-P-46843. There shall be a solder fillet between the board and the lead on the side of the board opposite the components. The fillet shall not obscure the contour of the clinched component lead. A solder fillet is acceptable, but not required, on the component side of the board.

3.3.1.7 Plated-through connections.- All plated-through connections shall be filled with solder in order to eliminate dependence on plating itself for circuit continuity.

3.3.1.8 Electrical clearance.- A minimum of .010 inch shall be maintained between uncommon component leads, transistor cases and tabs.

3.3.1.9 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 pounds per square inch (psi). The temperature of the cleaning solution shall not exceed 125 degrees Fahrenheit (°F). After cleaning there shall be no residual flux, surface contamination or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.1.10 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than the room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns.
b. Permissible moisture content limited to 2.12 grams per cubic foot (dew point 32°F maximum).
c. Dust count shall not exceed 1000 particles per cubic foot.

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3.3.1.11 Coating compound.

3.3.1.11.1 Masking.- Prior to coating, the areas specified on drawing F10547210 shall be masked. Subsequent to coating, with masking removed, all specified areas shall be clean and free of coating.

3.3.1.11.2 Coating.- The assembly shall be coated with Insulating Compound, Electrical, Type PUR, Polyurethane Resin conforming to MIL-I-46058.

3.3.1.11.3 Thickness.- Multiple applications shall be used with proper curing between each coat, until a minimum of .001 inches thick to .006 maximum thickness is achieve.

3.3.1.11.4 Cured compound.- The cured compound shall present a smooth continuous surface, and shall pass from the mounted components to the circuit board with no break at their junctions. The cured compound shall meet the following requirements:

a. Compound shall be free of visible dust particles or other foreign matter.
b. It shall be smooth and uniform and have no globules or bare spots.
c. It shall be substantially free of bubbles.
d. There shall be no blisters or cracks.
e. It shall be transparent and shall not affect marking legibility.
f. It shall be capable of being removed in local areas for replacement of components by means of a soldering iron or similar method.
g. Applied to a test board of 1/16 inch thickness, the compound shall not rupture or separate from the board when the board is bent into an arc of twelve inches radius.

3.4 Vibration.- The assembly shall withstand vibrations for 15 minutes in each of 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibrations shall be .036 inch at lower frequencies. Subsequent to vibration the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.5 Shock.- The assembly shall withstand a total of 6 shock impulses, one in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 millisecond. The peak amplitude of each shock impulse shall be 75 G's. Subsequent to shock

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the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.6 Storage temperature.- The assembly shall withstand thermal stabilization at temperatures -80°F and +160°F. Subsequent to exposure to the temperatures specified herein the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.7 Performance.

3.7.1 Operating temperature.- The assembly shall meet the requirements of 3.7.2 when thermally stabilized at all temperatures from -65°F to +155°F inclusive.

3.7.2 Operation.- The assembly shall function as specified in 3.7.2.1 to 3.7.2.5 with the specified supply voltages applied to the following pins:

Pin c +5.50 ± 0.30 VDC
Pin Y +4.0 ± 0.2 VDC
Pin D +3.50 ± 0.2 VDC
Pin W +4.0 ± 0.2 VDC (MEM) (Separate power source from that applied to Pin Y).
Pin 1 Power ground

3.7.2.1 High frequency counters(ON).- With a logical zero level (0.1 ± 0.1 VDC) applied to Pin n, the following outputs shall appear (see Figure 1):

Pin F - Sinusoidal Oscillation at 7.5 Mhz ± 0.075 Mhz, 1.4 ± 0.2 VDC (peak to peak).
Pin H - Positive pulses with an amplitude of 1.4 ± 0.2 VDC with a pulse width of 130 ± 10 ns repeated every $667\ ^{-0}_{+15}$ ns.

3.7.2.2 High frequency counters(OFF).- With a logical one level (1.1 ± 0.1 VDC) applied to Pin n, a DC level shall be established at Pins F and H. This DC level can be either a positive potential (1.1 ± 0.1 VDC) or ground potential (0.1 ± 0.1 VDC).

3.7.2.3 Reset.- With the logical one level (1.1 ± 0.1 VDC) applied to Pin n, momentary application of a +0.2 ± 0.2 VDC level to Pin o, shall establish the following DC levels:

Pin F = 0.1 ± 0.1 VDC Pin J = 1.1 ± 0.1 VDC
Pin H = 0.1 ± 0.1 VDC Pin L = 0.1 ± 0.1 VDC

3.7.2.4 Units numerical driver outputs.- Pin J and Pin L are complementary functions which shall be capable of acquiring alternate DC levels of 1.1 ± 0.1 VDC or 0.1 ± 0.1 VDC when Pin n is activated as in 3.7.2.2. When all operating voltages except the 4.0 VDC (MEM) on Pin W are removed the outputs at Pin J and Pin L shall maintain the DC voltage level acquired prior to the removal of the voltage.

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3.7.2.5 Decade numerical driver outputs.- The numerical indicator driver output terminals Q, G, O. E, M, C, A, N, B, P shall be capable of switching from an 80 ± 10 VDC level to a 1.0 ± 0.5 VDC level to indicate the state of the decade counter of the assembly. At any instant of time only one terminal shall be at the lower DC level with the other terminals maintaining the higher DC level. With all voltages removed except the +4.0 VDC (MEM) on Pin W, these output terminals shall maintain the voltage levels acquired prior to the removal of the voltages.

3.8 Electrical continuity.- The electrical wiring of the assembly shall conform to the electrical schematic drawing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplied and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected

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by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP)listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 1.0% DEFECTIVE			
101.	SPACING OF PARTS	3.3.1.1	SME; VISUAL
102.	SYMBOLS AND MARKINGS	3.3.1.2	VISUAL
103.	TRANSISTORS	3.3.1.3	VISUAL
104.	BEND RADIUS	3.3.1.4	SME; VISUAL
105.	COMPONENT LEADS	3.3.1.5	SME; VISUAL
106.	SOLDERING	3.3.1.6	VISUAL
107.	PLATED-THROUGH CONNECTIONS	3.3.1.7	SME; VISUAL
108.	ELECTRICAL CLEARANCE	3.3.1.8	VISUAL
109.	MASKING	3.3.1.11.1	VISUAL
110.	CURED COMPOUND	3.3.1.11.4	SME; VISUAL

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MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +70° and + 80°F.

TABLE II - CLASSIFICATION OF DEFECTS
100% INSPECTION

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR:		
112. PERFORMANCE	3.7 THROUGH 3.7.2.5	4.6.5 THROUGH 4.6.5.1.4
113. ELECTRICAL CONTINUITY	3.8	4.7
MINOR: NONE		

NOTE: The tests in Table II shall be conducted at a temperature between +70° and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.9
Drying	3.3.1.10
Coating, insulating compound	3.3.1.11

4.4 Special sampling.

4.4.1 General.- Three assemblies shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable Dwgs & Specs - visual

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TABLE III, CONT'D

302.	Fabrication	3.2	Applicable Dwgs & Specs - visual
303.	General specifications	3.3	MIL-F-13926, MIL-P-46843 - Visual

NOTE: The tests in Table III shall be conducted at a temperature between +70° and +80°F.

4.4.2 Environmental.- Three assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests of Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration (+70°F and +80°F)	3.4	4.6.1
305.	Shock (+70°F and +80°F)	3.5	4.6.2
306.	Storage temperatures -80°F and +160°F.	3.6	4.6.3
307.	Operating temperatures (-65°F and +155°F)	3.7.1	4.6.4

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2.2 Government design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed

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90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6:

	Nomenclature	Description
1.	D.C. Power Supply (4 Required)	Range of 0-10V, current rating 500 ma
2.	D.C. Power Supply	160 ± 5 VDC, current rating 10 ma.
3.	Oscilloscope	Risetime 10 nanoseconds or less, input impedance of 10 meg ohms or greater, sweep time of 100 nanoseconds per centimeter, sensitivity of .05 volts per centimeter.
4.	Pulse Generator	Sufficient capability to generate a manually controlled single positive 2 volt pulse (ground reference). Output impedance of generator shall be less than 50 ohms.

4.6 Test methods and procedures.

4.6.1 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The assemblies shall be firmly mounted to the vibration table. The vibration test shall then be performed in accordance with Test Method 514 of MIL-STD-810, except the vibration equipment shall be capable of providing the conditions of amplitude, frequency and duration specified in 3.4. Subsequent to vibration, the assemblies shall be subjected to a visual examination for missing or damaged parts and any physical damage caused by faulty workmanship or use of sub-standard material.

4.6.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926. The shock pulse shall comply with the waveform, time duration and peak "G" value specified in 3.5. Subsequent to the shock test, the assemblies shall be subjected to a visual and tactile examination to detect defects or damage to the assemblies.

4.6.3 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

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The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperatures specified in 3.6. At the conclusion of this test the assemblies shall be visually examined for any deterioration or damage to components and materials and then subjected to the tests of 4.6.5 through 4.7 at a temperature between 70° and 80°F.

4.6.4 Operating temperatures.- Conduct the performance test 4.6.5 in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.7.1, utilizing the testing equipment indicated in 4.5.3. The assemblies shall be exposed to each specified temperature for a period of 2 hours. At the end of each 2 hour period and while still at the specified temperature, the assemblies shall be subjected to the tests in 4.6.5 through 4.6.5.1.4.

4.6.5 Performance.- Connect the positive side of the +5.5 VDC, +4 VDC (MEM) +4 VDC (OSC), and +3.5 VDC power supplies (Item 1 in 4.5.3) to Pins c, W, Y and D respectively of the Assembly Connector P1. Connect the power supply grounds to Pin 1 of Connector P1. The above connections shall remain for all subsequent tests for compliance with 3.7.2, unless otherwise specified.

4.6.5.1 Operation.- Connect the test circuitry as shown in Figure 3.

4.6.5.1.1 High frequency counters.- Set Manual-Auto Switch to Auto-Position. Apply a ground potential to Pin n. The outputs F and H shall be as specified in 3.7.2.1. Apply 4 VDC through a 3.3k ohm resistor to Pin n. Pins F and H shall attain the DC levels as specified in 3.7.2.2.

4.6.5.1.2 Reset.- Set Manual-Auto Switch to Manual Position. Momentarily apply 4 VDC through a 470 ohm resistor to Pin o. The output at Pin L shall be +0.1 ± 0.1 VDC and Pin J shall be at a 1.1 ± 0.1 VDC level. Remove all the supply voltages from the assembly except the +4 VDC applied to Pin W. Pins L and J shall maintain the potentials they had prior to the removal of the voltages.

4.6.5.1.3 Units numerical driver outputs.- Re-apply the DC supply voltages to the appropriate pins of the assembly. While monitoring the output at Pin J, remove and re-apply the +4 VDC to Pin n through a 3.3k ohm resistor. Repeat this procedure until a zero DC level (0.1 ± 0.1 VDC) is established at Pin J. When this condition is established, remove all voltages except the 4V Memory from the board under test. The zero level established at Pin J prior to the removal of the voltages shall be maintained and the condition at Pin L shall be a DC level of +1.1 ± 0.1 VDC.

4.6.5.1.4 Decade numerical driver outputs.- To determine the performance of the decoding matrix and the drivers controlling the even digits of the numerical indicator tube, the following procedure is repeated five times, once for each of the even digits (0, 2, 4, 6 and 8). Set Manual-Auto switch to Manual Position. Turn off the +5.5 VDC supply at Pin c. Reset the circuits by momentarily applying +4 VDC through a 470 ohm resistor to Pin o. A zero shall be displayed on the numerical indicator after each reset. Apply the output of the manually controlled single pulse generator, through the external

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circuit shown in Figure 2, to Pin F. The digit displayed will be twice the number of pulses applied to Pin F. After insertion of the desired number of pulses, remove the +3.5 VDC from Pin D and the +4 VDC from Pin Y. The digit displayed prior to removal of the above voltages shall remain displayed. Re-apply the above voltages and repeat the sequence for all the even digits. To determine the performance of the memory, decoding matrix, and the drivers controlling the odd digits of the numerical indicator, the following procedure is repeated five times, once for each of the odd digits (1, 3, 5, 7 and 9). Turn the +5.5 VDC supply on and off randomly until an odd digit is displayed on the numerical indicator. Leave the +5.5 VDC power supply turned off. Apply the output of the single pulse generator to Pin F. The digits displayed shall progress in sequence of 1, 3, 5, 7 and 9. When the desired digit is displayed, remove the +3.5 VDC from Pin D and the +4 VDC from Pin Y. The digit displayed prior to the removal of the above voltages shall remain displayed. Re-apply the voltages and repeat the sequence for the remaining odd digits.

4.7 Electrical continuity.- Continuity shall be inspected by standard measuring equipment (SME) and visually examined for compliance with Drawing F10547210 as specified in 3.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10547210. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Circuit Card Assembly, Oscillator, Quinary, Memory and Tens Matrix - A1 generates the basic timing signal used for time interval measurement. It is a subassembly of the Counter, Time Interval which is a module of the Range Finder, Laser, AN/GVS-3.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable packaging data sheet.
c. Selection of applicable levels of preservation, packaging and packing.
d. Provisions for first article testing.

Custodian:

Army - MU

Preparing activity:

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Project No. 1240-A197

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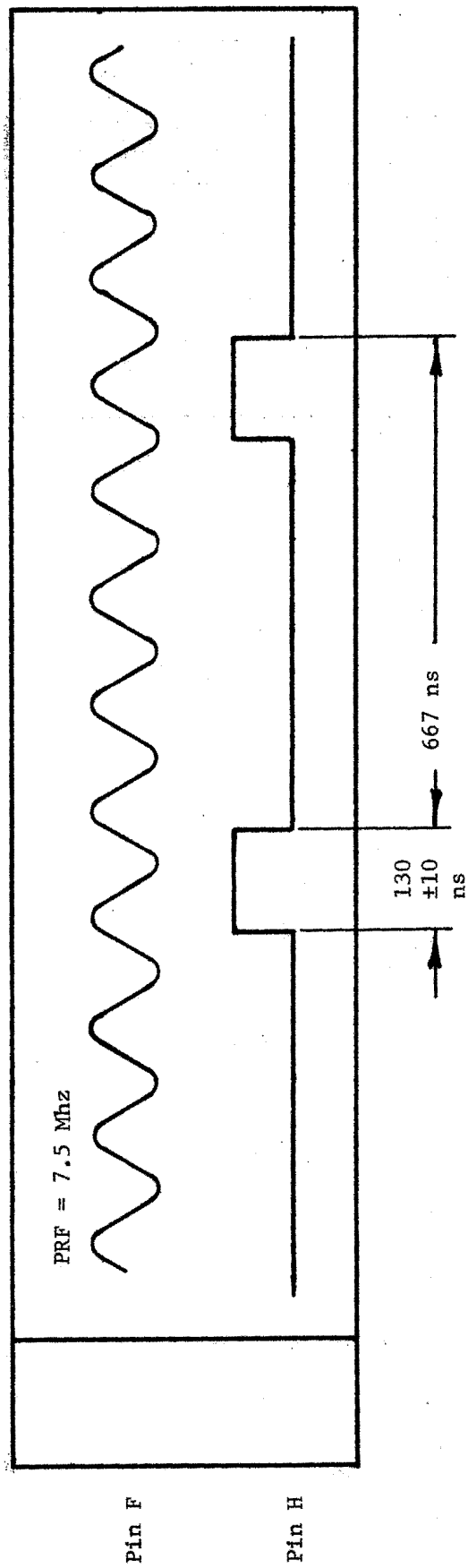


FIGURE 1

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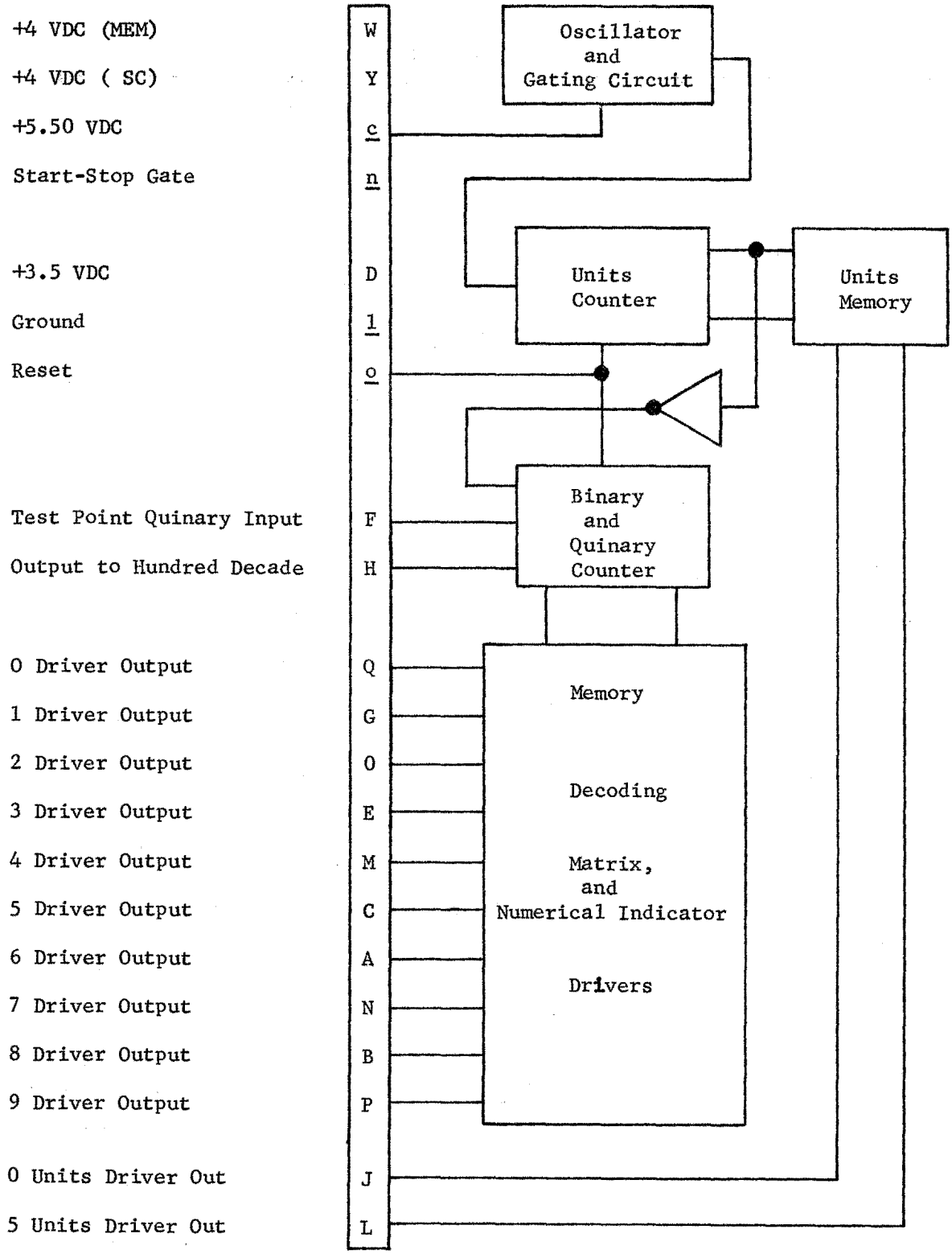


FIGURE 2

FUNCTIONAL BLOCK DIAGRAM

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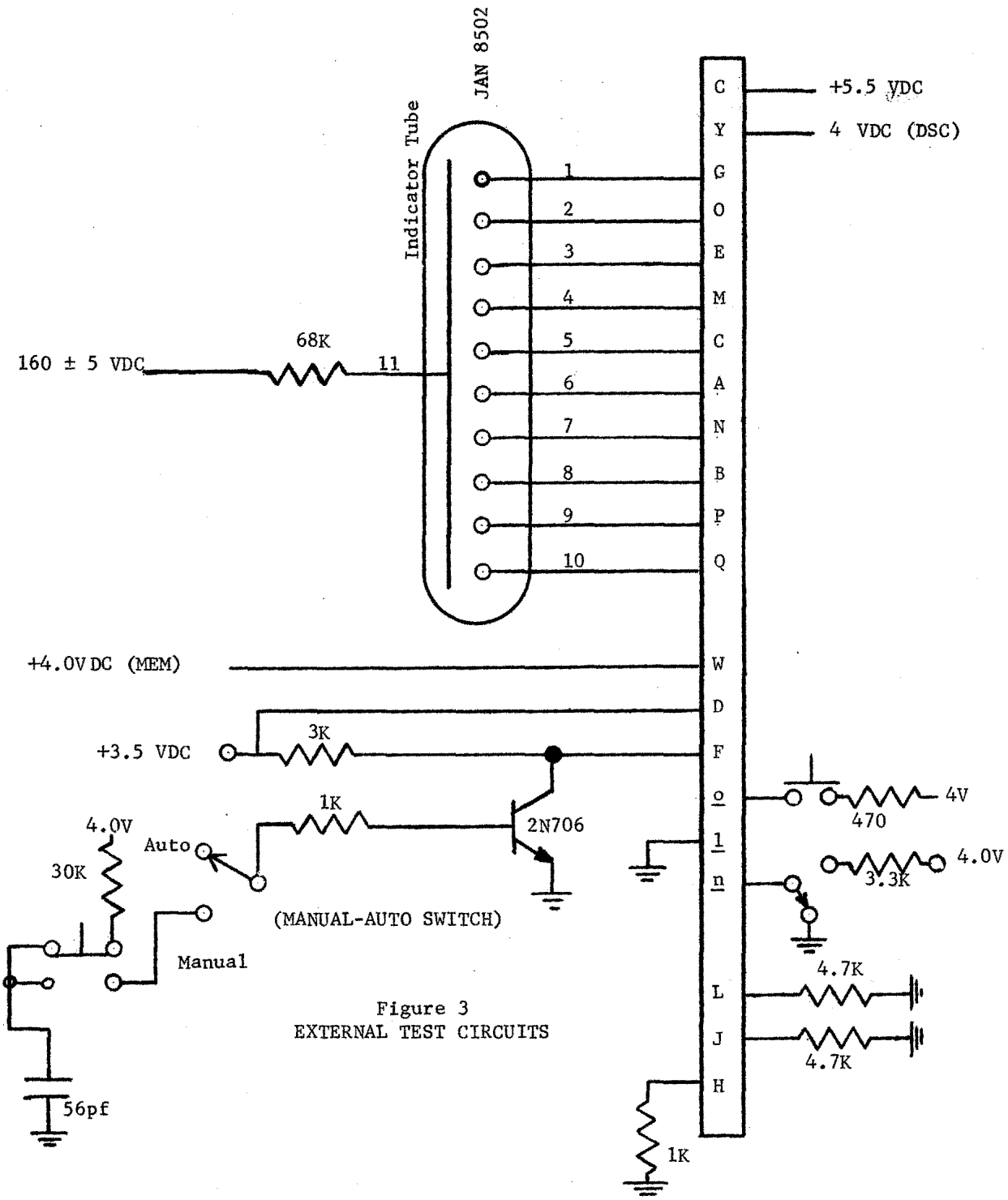


Figure 3
EXTERNAL TEST CIRCUITS

All Resistors 1/4w, ± 5%

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